Texas A&M University at Qatar

Electrical and Computer Engineering Program

ECEN 403

Rakib - AI Helmet for Safety

Benchmarking

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"On my honor, as an Aggie, I have neither given nor received unauthorized aid on this academic work."

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1. Introduction

Problem Statement

Fatigue is something riders in Qatar face a lot especially during extreme hot weather and long working shifts. Long shifts without rest can lead to accidents and delayed emergency response. Traditional conventional helmets provide only passive protection during impacts and do not help prevent accidents. Available smart helmets focus mainly on communication or navigation features. These products add value to the market but they do not provide proactive fatigue monitoring or automatic emergency response for riders. The objective of this project is to develop a smart helmet that enhances rider safety by detecting fatigue and crashes early, thus preventing accidents and providing critical post ride information.

Proposed Design

The Rakib AI Helmet is designed to keep delivery riders safe by using AI and smart sensors to detect fatigue and crashes in real time. It uses a MAXM86161 (PPG/HRV) sensor, and MPU-6050 (Accelerometer/gyroscope) to monitor the rider's heart rate variability and impact for reliable fatigue and impact detection.

The LilyGO T-SIM7600G-S3 (ESP32) microcontroller processes the collected data in real time and when the system is offline, fatigue alerts are provided locally through the LED indicators while the AI assistant becomes active through the connected communication system when the helmet is online. In the event of a crash, the LTE/GNSS module automatically sends an SOS alert with the rider's live location to emergency services and then later on stored in the Rakib mobile app. Post ride data is uploaded in the Firebase cloud and mobile app and enables users and fleet managers to continuously analyze fatigue patterns and riding behavior.

The Rakib Helmet not only protects riders but also prevents accidents; it is a proactive advancement toward safer delivery operations in Qatar.

2. Existing Solutions

1) Quin Smart Helmet

The Quin Design smart helmet provides emergency warnings, crash detection, and Bluetooth connectivity through a linked smartphone app [1]

- Advantages: Include hands-free communication, an appealing design, and embedded sensors for impact detection.
- Disadvantages: Its lack of physiological fatigue sensing and heavy reliance on smartphone connectivity are its drawbacks.
- Constraints: It is not AI-driven and is only compatible with smartphone-based SOS.

2) Sena Smart Helmet

Sena helmets prioritise multimedia and intercom capabilities for riders. Voice control and optional crash sensors are available on certain model [2]

- Advantages: Include a dependable intercom, a great communication range, and group riding convenience.
- Disadvantages: It has limited safety analytics, uses more power, and is incapable of sensing weariness
- Constraints: It does not integrate with the cloud and is primarily intended for non-professional riders

3) Forcite MK1 Helmet

Forcite does not track physiological measures or fatigue, but it has cameras, navigation lights, and effective communication. [3]

- Advantages: It features a built in HD camera, and it is made from strong and high quality materials.
- Disadvantages: SOS must be activated manually, and no AI or fatigue models are available
- Constraints: It only targets the higher end market and has a high production cost

4) LIVALL Smart Helmet

LIVALL helmets pair with smartphones to provide Bluetooth alerts and fall detection [4]

- Advantages: They include integrated fall alerts and communication, are reasonably priced, and are lightweight
- Disadvantages: They lack fatigue data, have restricted AI processing, and have poor impact detection accuracy
- Constraints: They have to always be linked to the smartphone app

5) Proposed Rakib AI Helmet

- Advantages: It combines fatigue and crash detection, mobile app /cloud-based SOS, and analytics.
- Disadvantages: The initial cost of the prototype is higher than that of commercially available helmets, as well as AI model training and dataset collection is needed.
- Constraints: System performance is dependent on LTE and GNSS coverage for reliable SOS transmission.

3. Design Constraints

Power Limitation: Continuous Sensor operation, and data transmission battery life is depleted. To ensure that performance is kept at a high standard and safety is protected optimization of the battery and model is needed.

Environmental Factors: Must withstand unpredictable and extreme weather conditions. Such as Qatar's high temperatures and humidity to ensure durability.

Data Accuracy: Physiological sensors must maintain signal stability under motion and are essential for accuracy.

Connectivity: LTE/GNSS availability may limit SOS reliability and real time transmission.

Cost: Prototype cost target below USD 1000 for scalability and market entry potential.

4. Benchmarking Criteria

Benchmarking evaluates how Rakib and existing products perform under engineering and societal factors, including public health and welfare criteria while taking into account environmental and economic considerations:

Criterion	Description
Public Health	Enhances rider safety by preventing fatigue-related accidents and crash detection.
Public Safety	Ensures timely emergency alerts and secure data handling.
Public Welfare	Provides riders with confidence, comfort, and reliability .
Cultural & Social Factors	Adapts to Qatar's delivery culture; supports local working conditions.
Global & Environmental Factors	Promotes road safety sustainability; reduces injury-related social cost.
Economic Factors	Evaluates affordability, scalability, and market adoption potential.

5. Benchmarking Table

Feature / Metric	Quin Helmet	Sena Helmet	Forcite MK1	LIVALL Helmet	Rakib AI Helmet (Proposed)
Fatigue Detection (HRV/IMU)	NA	NA	NA	NA	A
Crash Detection (AI-Based)	A (via Smartphone)	A	A	A (basic)	A (on-device)
Automatic SOS (w/ GPS)	A (phone required)	NA	NA	A	A (LTE/GNSS)
On-Device AI Processing	NA	NA	NA	NA	A (ESP32 + TensorFlow Lite)
Data Analytics / Cloud Storage	NA	NA	NA	NA	A (Firebase + App)
Voice / User Interface	A	A	A	A	A
Public Health Impact	Moderate	Moderate	Low	Moderate	High

Power Consumption	Medium	High	High	Low	Medium (optimized)
Cost (Approx. USD)	700	600	1000	200	900
Availability	USA	Global	Australia	Global	Local Prototype (Qatar)
Overall Safety Level	Medium	Medium	Medium	Low	Very High

NA - Not Available A - Available

The benchmarking results indicate that the Rakib AI Helmet provides superior safety performance through AI-based fatigue and crash detection compared to existing helmets and utilizes the mobile app efficiently for real time monitoring, alerts and data visualization. While the prototype cost is moderately higher, its added intelligence, LTE-based SOS system, and data analytics capabilities justify the trade-off. Future improvements will focus on power optimization and expanding connectivity support to ensure consistent performance.

6. Benchmarking Study Analysis and Summary

The purpose of this benchmarking study was to understand how existing smart helmets work and to find ways to make the design better. After comparing Rakib with helmets such as Forcite, Sena, LIVALL and Quin it became clear that most of them only react after a crash happens. The main idea of the project, Rakib, focuses on preventing accidents by detecting fatigue early and sending alerts before a crash.

This study includes several useful ideas that will help improve the design. For example, to use lightweight materials like LIVALL to make the helmet more comfortable, and also apply Sena's battery management methods to improve power life. As well as Quin's mobile app design inspired improvements to make the Rakib app more user friendly, while Forcite's padding and heat control inspired riders' comfort.

Overall, the benchmarking process made the Rakib design clearer and helped in differentiating Rakib by combining AI-based fatigue detection, crash alerts and SOS communication all in one helmet.

7. Conclusion

The Rakib Helmet project is headed in the right direction, according to the benchmarking process. Using AI and built-in SOS features, Rakib differentiates itself from other helmets by highlighting both prevention and response. The findings showed that the concept is feasible, necessary, and highly beneficial, especially for delivery riders in Qatar.

The study also helped improve the system's design, comfort and power management. Future work will focus on making Rakib more energy-efficient, collect more real-world data to improve the AI models and also move toward certification so it can reach the market as a safe and reliable helmet.

8. References

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